

Unit 2 — Exponents & Scientific Notation

★ Answer Key ★

Do not distribute to students until after the lesson.

Guided Notes — Fill-Ins

2.1 Exponent Rules. Product: $x^3 \times x^4 = x^7$; $(2x^3)(3x^2) = 6x^5$; “Same base? ADD.” **Quotient:** $x^6 \div x^2 = x^4$; $(12x^7) \div (3x^2) = 4x^5$; “Same base? SUBTRACT.” **Power:** $(x^2)^3 = x^6$; $(2x^3)^2 = 4x^6$; “Power to power? MULTIPLY.” **Zero:** $5^0 = 1$; $(3x^5)^0 = 1$; “ANY $\text{base}^0 = 1$.” **Negative:** $2^{-3} = 1/8$; $4x^{-3} = 4/x^3$; “Flip to a fraction.” Parentheses: $-3^2 = -9$ vs $(-3)^2 = 9$.

Quick Check: Product Rule $\rightarrow x^5 \cdot x^3 = x^8$. **Match:** Product $\rightarrow$ “Same base? ADD”; Quotient $\rightarrow$ “Same base? SUBTRACT”; Power $\rightarrow$ “Power to power? MULTIPLY”; Negative $\rightarrow$ “Flip to a fraction.” **Worked:** $(4x^3y^2)(3x^5y) = 12x^8y^3$. **Error:** $(3x^2)^3 = 27x^6$.

2.2 Scientific Notation. Table: Large number $\rightarrow$ exponent + (positive); Small number $\rightarrow$ exponent – (negative). **Multiply:** MULTIPLY coefficients, ADD exponents, adjust to between 1 and 10. **Divide:** DIVIDE coefficients, SUBTRACT exponents. Examples: $(3 \times 10^4)(2 \times 10^5) = 6 \times 10^9$; $(8 \times 10^6) \div (4 \times 10^2) = 2 \times 10^4$; $(5 \times 10^3)(4 \times 10^2) = 20 \times 10^5 \rightarrow 2.0 \times 10^6$. **Error:** $0.0045 = 4.5 \times 10^{-3}$.

Practice A — Exponent Rules with Monomials

1) x^8 2) y^{12} 3) a^5 4) $10x^5$ 5) $27m^6$ 6) $3y^5$ 7) $4a^8$ 8) $15x^3y^4$ 9) 3 10) $8b^5$ 11) $12x^5$ 12) $x^{15}y^6$

Practice B — Conversions

1) 4.5×10^7 2) 3.2×10^{-4} 3) 602,000,000 4) 0.000091 5) 7.8×10^5 6) 6.7×10^{-6} 7) 34,500 8) 0.0012 9) 5.28×10^9 10) 0.025

Practice C — Operations

1) 6×10^9 2) 2×10^4 3) 3.5×10^{10} 4) 3×10^3

Practice D — Word Problems

1) $(9.3 \times 10^7) \div (1.86 \times 10^5) = 5 \times 10^2$ s (500 s) 2) $(2 \times 10^{-6}) \div (5 \times 10^{-8}) = 40 \times$ longer 3) $(3.4 \times 10^{13}) \div (3.4 \times 10^8) = 1 \times 10^5 = \$100,000$ 4) $(6.5 \times 10^{-4})(8 \times 10^{18}) = 5.2 \times 10^{15}$ g 5) $(3.7 \times 10^{13})(1 \times 10^{-5}) = 3.7 \times 10^8$ m 6) $(1.44 \times 10^{12}) \div (4.8 \times 10^9) = 3 \times 10^2$ s (300 s)

Cumulative Review

A • Exponent Rules: 1) x^{10} 2) m^{12} 3) p^5 4) $12a^7$ 5) $-8c^{12}$ 6) $3x^3y^2$

B • Conversions: 1) 6.3×10^6 2) 7.1×10^{-4} 3) 208,000 4) 0.0000045 5) 9.1×10^7 6) 0.036

C • Operations: 1) 8×10^9 2) 3×10^6 3) $30 \times 10^9 \rightarrow 3.0 \times 10^{10}$ 4) 4×10^4

D • Word Problems: 1) $(8 \times 10^{-6}) \div (2 \times 10^{-6}) = 4 \times$ wider 2) $(1.5 \times 10^8) \div (3 \times 10^4) = 0.5 \times 10^4 = 5 \times 10^3$ h (5,000 h) 3) $(5 \times 10^3)(2 \times 10^2) = 10 \times 10^5 = 1 \times 10^6$ widgets